Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Re: Subject: Selection Functionals Paper - My Response to Your Latest Comments

Ezequiel Boero <ezequiel.boero@adagioinstitute.org> Tue, Sep 29, 2026 at 10:24 AM
To: "Benjamin D. Summers" <bsummers@adagiollc.com>
Cc: "ISMAIL-SUTTON, SARA R." <sara.r.ismail-sutton@durham.ac.uk>

Ben,

Thanks for the reply and the check. You are right; my sweep violated the proof's inequalities, not the proposition's conclusion.
I wrote the second when I had only shown the first. Convergence in Q-probability needs $\mathbb{E}[V_M] \to \sigma_\infty^2 T$
and $\operatorname{Var}(V_M) \to 0$, and neither requires $E \succeq 0$.

Rederived, your account holds. Over the surrogate family with $\sigma_\Pi = \sigma_\nu = 1$, $\lambda = 1/2$, $k_\nu \in \{0.5, 1, 2, 5\}$, $k_\Pi \in \{-1, -5, -20\}$ at $M = 1600$: the worst mean error is 0.0079 and $M\cdot\operatorname{Var}$ stays
between 0.871 and 0.889. Your extreme case reproduces exactly: 0.607872 and 0.650976 at $M = 200, 800$ against your 0.6079 and 0.6510,
error ratio 3.75; my third grid was 1600 rather than 3200, ratio 1.98 for the twofold step.

It holds outside the family too, in the worst case for the bounds: with $k_\Pi = -3$, $k_\nu = -2$, $\sigma_\nu = 1.3$ the mean converges
to 0.771689 with errors -0.0886, -0.0225, -0.0113, and $M\cdot\operatorname{Var}$ falls from 5.74 to 1.78.
And your structural argument checks: by Corollary 3.6(ii), $a_{\mathrm{eff}} \in (0,1]$ for either sign (0.9899 in the extreme case, 0.9983 with both rates negative) so the selected chain is stable and its innovation variance tends
to $\sigma_\infty^2\delta t$ whatever the sign of κ.

So $\kappa \geq 0$ is a hypothesis of that proof and not of the result: the proposition is stronger than its proof.
The manuscript already says this correctly.

Best,
Ezequiel

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